

Prisha Priyadarshini

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EDUCATION

Rutgers University – New Brunswick

Bachelor of Science in Computer Science, Minor in Mathematics GPA: 3.50/4.00

New Brunswick, NJ

Expected May 2027

RESEARCH EXPERIENCE

Lead ML Research Extern / Rutgers MBS Exchange

Jan. 2026 – Present

The Center for Discrete Mathematics & Theoretical Computer Science (DIMACS)

New Brunswick, NJ

- Led an empirical study under Dr. Linda Ness comparing interpretable decision tree models (**SPLIT**, **LicketySPLIT**, **LicketyRESPLIT**, **GOSDT**) with SOTA boosting methods (**XGBoost**, **CatBoost**, **LightGBM**) across 6 datasets.
- Demonstrated that preprocessing with **ThresholdGuessBinarizer** eliminates performance gaps between boosting models and interpretable decision trees, achieving comparable accuracy while preserving interpretability.
- Analyzed the impact of preprocessing techniques (**SMOTE**, **TGB**) on **Rashomon set size**, showing that feature binarization significantly expands model equivalence classes.
- Quantified **performance–complexity tradeoffs** using accuracy, recall, runtime, and Rashomon set size, finding that increased model complexity yields marginal performance gains but substantially larger Rashomon sets.

AI Researcher (Multi-Agent Deliberation & Consensus Dynamics)

Sept. 2025 – Present

Algoverse AI

Remote

- Built a multi-agent framework to study consensus formation, showing that **model-to-model deference drives convergence over independent reasoning**.
- Ran large-scale 20-round deliberation experiments across **GPT-4.1**, **Mistral**, **LLaMA**, and **Phi** models on subjective and objective benchmarks.
- Introduced quantitative metrics (Inter-agent disagreement rate, model deference rate) to analyze interaction dynamics, identifying **hierarchical influence patterns (up to 80% small to large model deference)**.
- Developed a rotation-based framework to separate model identity from response quality, revealing that **large-to-small model deference rates increase relative to baseline**, slightly weakening hierarchical influence patterns.
- Showed that prompting strategies can control multi-agent behavior, **increasing disagreement and reducing deference, with implications for system reliability**.
- In submission to 2 ICML workshops (Pluralistic Alignment and AI4GOOD).

AI Researcher (Multimodal Machine Learning)

June 2025 – Dec. 2025

Algoverse AI

Remote

- Contributed to a hierarchical scene-captioning pipeline **integrating dynamic stride window selection and multimodal chain-of-thought (MMCoT) reasoning** for temporally coherent caption generation.
- Designed and implemented evaluation frameworks (BLEU-4, METEOR, CIDEr, BERTScore) and conducted ablation studies on frame sampling strategies.
- Ran large-scale YouCook2 experiments on 30 RTX A6000 GPUs and achieved **17% higher CIDEr than GPT-4o** and **14% over VLLaMA-3**.
- Accepted for an **oral presentation at NeurIPS 2025 (7HVU)** and **accepted to AAAI 2026 (AI4EDU)**.

Algorithms Research Shadow

Jan. 2025 – May 2025

The College of New Jersey

Ewing, NJ

- Researched classical and modern folding algorithms under Dr. Dimitris Papamichail (Nussinov, Zuker, LinearFold, MCTS).
- Implemented sparse dynamic programming to prune redundant structures and reduce runtime.
- Conducted SLURM-based large-scale experiments using ViennaRNA across thousands of RNA sequences.

PUBLICATIONS

• Directional Influence and Consensus Formation in Multi-Agent Systems

Priyadarshini, P., Shrivastava, A.

In submission to ICML AI4GOOD and Pluralistic Alignment workshops

• DynaStride: Dynamic Stride Windowing with MMCoT for Instructional Multi-Scene Captioning

Pham, E., Priyadarshini, P., Maliackel, A., Bandi, K., Meo, C., Zhu, K.

NeurIPS 2025 7HVU (Oral), AAAI 2026 AI4EDU

INDUSTRY EXPERIENCE

Incoming Machine Learning InternMay 2026

RegeneronTroy, NY

- Incoming Summer 2026

AI/ML Open Source Contributor (DeepChem)May 2026

Google Summer of CodeRemote

- Selected as **1 of 5 out of 200+ applicants** for DeepChem to integrate **OLMo 7B models** into its LLM infrastructure
- Extending the **HuggingFace** wrapper with **generation, batching, and causal LM support**, enabling scalable training and inference for large decoder-only models such as OLMo.

AI Associate DeveloperOct. 2025 – Jan. 2026

Insurity – SpatialKey TeamRemote

- Built **geospatial preprocessing pipelines over 6M+ records**, integrating climate and peril data via time alignment and spatial filtering.
- Engineered features using Pandas and GeoPandas to support ML models.
- Built and evaluated LightGBM-based multi-class peril classifiers; **applied SMOTE class balancing, increasing accuracy and F1 score by 10%.**
- Experimented with a Temporal Fusion Transformer (TFT) model, modeling seasonal dependencies via cyclical week encoding and trained GPU-accelerated PyTorch models in a managed Anaconda environment on NVIDIA RTX hardware.
- Additionally, developed a geospatial change-detection system for before/after wildfire satellite imagery using OpenCV and U-Net++ for high-resolution segmentation.

PROJECTS

Knowledge Distillation for Reasoning in MoE and Dense LLMs | *Python, PyTorch, HuggingFace* April. 2026

- Developed an offline knowledge distillation pipeline using **LoRA (< 1% trainable parameters)** to transfer chain-of-thought reasoning and token-level distributions from large LLMs to a 3B student via **sequence and logit distillation (KL divergence + temperature scaling)**, achieving **4.5% accuracy gains** and **96%+ BERTScore on SciBench and TheoremQA.**

GenreBlender | *Python, PyTorch, Streamlit* Feb. 2026

- Engineered a Generative AI system combining Meta’s MusicGen and a **4-layer PyTorch Multilayer Perceptron (92% validation accuracy)** to create and quantitatively evaluate controllable music genre blends via a weighted probability framework in a Streamlit app.

PocketRAG | *Python, Flask, FAISS, Gemini, AWS, Docker, RAG* Sept. 2025

- Engineered a production RAG pipeline with document chunking, **FAISS vector retrieval**, and Gemini inference; containerized and deployed on **AWS EC2 (Docker + Gunicorn)** achieving **sub-second latency.**

PROFESSIONAL DEVELOPMENT & EXTRACURRICULARS

Mathematics GraderRutgers University – New Brunswick

Dec. 2025 – Present

- Grade homework and exams for Multivariable Calculus across 3 sections

Peer TutorThe College of New Jersey

Sept. 2024 – May 2025

- Provided one-on-one tutoring for Data Structures, Algorithms, Linear Algebra, and Statistics

Recruitment ChairKappa Theta Pi

Dec. 2024 – May 2025

- Led recruitment efforts and onboarded 10 new members for Spring 2025

Student Athlete (NCAA Division III Tennis)The College of New Jersey

Aug. 2023 – May 2024

- Balanced 20+ hrs/week training with academics; earned All-NJAC recognition

TECHNICAL SKILLS

AI & ML: PyTorch, Hugging Face, OpenCV, Transformers, scikit-learn

Research Domains: Multi-agent Systems, Interpretable ML, Multimodal ML

Programming Languages: Python, C, C++, JavaScript, Java

Libraries: NumPy, Pandas, GeoPandas, Matplotlib, Seaborn

Cloud, Systems, & MLOps: GitHub, AWS, Azure, Docker, CI/CD, pytest, Jira, Anaconda, Vercel

Certifications: AWS Certified Machine Learning — Specialty, AWS Certified Cloud Practitioner